

PBR-H95Q

Mixed-Precision Base + Sparse Precision Recovery

PoC Implementation and Evaluation Guide

Lossy preconditioning followed by exact entropy coding

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Research status: The real-Qwen lossless PBR experiments correctly reached the approximately seven-bit mantissa wall. Phase A of H95 established a deterministic and idempotent 7/5 policy near 9.29 total BPW. PBR-H95Q tests whether controlled mantissa reduction creates enough symbol, bit-plane, residual, and pair structure for exact post-quantization coding to approach 8 total BPW while retaining at least 95% of held-out model quality.

1. Purpose and Core Hypothesis

PBR-H95Q separates the experiment into two stages. Stage 1 deliberately reduces floating-point precision under a model-quality budget. Stage 2 losslessly compresses the resulting quantized checkpoint. The original BF16 checkpoint is not expected to reconstruct exactly; the quantized reference checkpoint must reconstruct exactly.

Core hypothesis: mantissa quantization reduces the active alphabet, forces low bit planes to zero, increases duplicate symbols, and may concentrate spatial residuals. Those changes can make fixed-width packing, rANS, Huffman, bit-plane coding, matrix residuals, and residual-pair dictionaries more effective than they were on the original dense BF16 mantissas.

```
Original BF16 -> quality-controlled quantization -> quantized reference
Quantized reference -> exact candidate coding -> independent decoder
Decoded quantized model == quantized reference
Model score(decoded quantized model) / score(original BF16) >= 0.95
```

2. Starting Point and Target

Item	Current value	Meaning
Lossless PBR bundle	~10.62 BPW	Sign ~1, exponent ~2.62, mantissa ~7.
Phase A mixed 7/5 policy	~9.29 BPW	Deterministic and idempotent; quality gate still requires held-out evaluation.
H95Q target	~8.00 BPW	Requires approximately 1.29 BPW additional reduction from the Phase A rate.
Available mantissa budget at 8 BPW	~4.38 BPW	Assumes sign ~1 and exponent ~2.62 before new metadata.

The target equation is:

$$8.00 - 1.00 \text{ sign} - 2.62 \text{ exponent} = 4.38 \text{ BPW}$$

Therefore, the weighted quantized mantissa stream plus all added precision maps, exception maps, dictionaries, and alignment should approach or beat 4.38 BPW.

3. Quantization Principles

- Precision reduction is the primary rate lever. Entropy coding is a secondary optimization.
- A K5 tensor has only 32 retained mantissa symbols; a K4 tensor has 16; a K3 tensor has 8.
- Removed low mantissa bits must not be serialized as zeros. Store only the retained bits or an entropy-coded equivalent.

- Quantization must use a deterministic rounding policy and explicitly handle mantissa carry into the exponent.
- The quantizer must be idempotent: $Q(Q(W)) = Q(W)$.
- Calibration, validation, and held-out evaluation data must remain separate.
- The 95% threshold refers to model behavior, not the percentage of unchanged weights.

4. Candidate Precision Modes

Mode	Retained mantissa bits	Fixed mantissa baseline	Intended use
K7	7	7 BPW	Sensitive tensors, control tensors, exact fallback.
K6	6	6 BPW	Intermediate protection when K5 is too aggressive.
K5	5	5 BPW	Conservative reduction; current Phase A family.
K4	4	4 BPW	Primary route toward approximately 8 total BPW.
K3	3	3 BPW	Robust tensors only after measured validation.

5. Four Complete PoC Candidates

Candidate	Precision design	Post-quantization coding	Purpose
A — H95 control	Existing per-tensor 7/5 policy	Packed retained bits	Reproduce the ~9.29-BPW Phase A baseline.
B — mixed precision	Per-tensor 7/6/5/4/3	Packed retained bits	Measure the gain from precision allocation alone.
C — sparse recovery	K4 base plus K5/K6/K7 protected weights or channels	Packed base + compressed exception map + corrections	Trade a small correction stream for better quality.
D — hybrid coded	Candidate C precision map	Best of packed, rANS, bit-plane, matrix residual, pair dictionary	Measure additional exact compression after quantization.

6. Candidate Baseline: Packed Retained Bits

Every quantized tensor must first be compared with a fixed-width packed baseline:

```

K7 tensor -> 7 bits per mantissa
K6 tensor -> 6 bits per mantissa
K5 tensor -> 5 bits per mantissa
K4 tensor -> 4 bits per mantissa
K3 tensor -> 3 bits per mantissa

```

No advanced codec may be selected unless its complete serialized cost, including mode identifiers and shared dependencies, beats this packed K-bit baseline.

7. Exact Coding Methods After Quantization

7.1 Quantized-symbol rANS or arithmetic coding

Measure the symbol entropy of the retained K-bit values. K5 has at most 32 symbols and K4 has at most 16. Use rANS only when the measured distribution is sufficiently non-uniform to beat fixed-width packing after table cost.

```
gain = packed_K_bits - complete_rANS_bits
```

7.2 Bit-plane coding

Transpose retained mantissas into bit planes and test each plane independently. Some planes may be strongly biased even when whole-symbol entropy is near K bits.

- RAW bit plane
- Run-length coded bit plane
- Context-coded bit plane
- rANS-coded bit plane

7.3 Matrix residual coding

Re-run LEFT, UP, PREVIOUS, AVG, and PAETH predictors only on quantized values. Quantization may convert small numerical differences into repeated values and zero residuals. Matrix mode remains optional and must beat packed K-bit RAW after complete cost.

7.4 Residual-pair dictionary

Pair two quantized residuals horizontally, vertically, or in traversal order. Encode frequent pairs with bit-packed dictionary IDs and use exact escapes for misses. Train dictionaries only on the calibration partition, freeze them, and charge their serialized cost.

7.5 Zero-run and sparse coding

If quantization creates long zero runs in residuals or bit planes, test run-length and sparse-index representations. Use delta-coded indices or compressed bitmaps only when their complete cost wins.

8. Sparse Precision Recovery

Candidate C begins with a low-precision base, preferably K4, and restores additional low mantissa bits only for salient weights, rows, channels, or blocks.

```
base_stream      = K4 for all selected weights
exception_map    = locations requiring extra precision
correction_stream = 1, 2, or 3 additional low bits
```

Illustrative arithmetic:

```
95% of weights at K4 + 5% with 3 correction bits
mantissa rate ≈ 4 + 0.05*3 = 4.15 BPW before map cost
total rate ≈ 1 + 2.62 + 4.15 = 7.77 BPW before metadata
```

This calculation is only a feasibility illustration. The actual exception-map cost and held-out quality determine whether the design succeeds.

9. Precision Allocation Policy

Tensor family	Initial H95Q candidates	Notes
Embedding / tied output	K7, K6, K5, then row tiers	Large target; existing calibration suggests K5 is mild, but held-out quality is required.
Normalization and small vectors	K7	Small storage contribution and potentially sensitive.
Attention projections	K7, K6, K5	Current proxy suggests mid-attention can be more sensitive than embedding.
MLP gate/up/down	K5, K4, selected K3	Primary candidate for aggressive reduction.
First and last blocks	K7, K6, K5	Protect initially, then relax only if measured safe.
Outlier channels or rows	Sparse K6/K7 recovery	Preserve selected high-impact regions

10. Sensitivity and Selection Score

For each proposed precision change, measure bytes saved and calibration loss:

```
utility = bytes_saved / max(delta_quality, epsilon)
select highest utility candidates first
periodically evaluate the combined policy because quantization errors interact
```

A candidate should be reverted when it causes disproportionate degradation in:

- NLL or perplexity
- Task accuracy or exact match
- Top-1 and top-5 logit agreement
- Numbers and code tokens
- Special/control-token behavior
- Multilingual subsets
- Safety or refusal behavior

11. Experimental Workflow

1. Freeze the original BF16 model revision, tokenizer revision, runtime, and evaluation code.
2. Measure the original BF16 quality baseline.
3. Reproduce Candidate A and its approximately 9.29-BPW rate.
4. Generate Candidate B using per-tensor K7/K6/K5/K4/K3 choices.
5. Create Candidate C using a K4 base and sparse high-precision recovery.
6. For every quantized tensor, generate Candidate D exact storage alternatives.
7. Decode every serialized candidate independently and compare it with the quantized reference.
8. Measure complete physical size, not estimated payload alone.
9. Evaluate validation quality and revise the precision allocation.
10. Run the untouched held-out evaluation once the policy is frozen.
11. Accept only configurations retaining at least 95% of baseline quality.

12. Required Measurements

Category	Required measurements
Precision	Weight count and percentage at K7/K6/K5/K4/K3.
Entropy	Retained-symbol entropy and entropy per bit plane.
Residuals	Zero-residual rate and $ \text{signed residual} \leq 1/\leq 2$ rates.
Pairs	Top-16/32/64/128/256 coverage and pair entropy.
Exceptions	Protected percentage, map bits, correction bits, and total exception BPW.
Storage	Sign, exponent, mantissa, maps, dictionaries, alignment, checksum, and total BPW.
Correctness	Differing quantized words, quantized-reference SHA-256, decoded SHA-256.
Quality	Baseline score, quantized score, retention, subgroup scores, and ΔNLL .
Runtime	Peak memory, load time, throughput, latency, and kernel compatibility.

13. Critical Diagnostic: Fixed Rate vs Actual Entropy Rate

For each tensor and precision, compare:

```

fixed_rate_BPW = K
measured_entropy_BPW = H(retained symbols)
actual_codec_BPW = complete serialized bits / weight count

```

Interpretation examples:

```

K5 fixed = 5.000, entropy = 4.72, rANS = 4.78 -> useful ~0.22-BPW gain
K5 fixed = 5.000, entropy = 4.99, rANS = 5.03 -> reject rANS; use packed K5

```

14. Correctness Gates

14.1 Quantizer gates

- ❑ Idempotence: $Q(Q(W)) = Q(W)$.
- ❑ Deterministic rounding and exponent-carry behavior.
- ❑ Documented NaN, infinity, subnormal, and signed-zero policy.
- ❑ Stable precision map for identical input and configuration.

14.2 Storage codec gates

- ❑ Decoded quantized words equal the quantized reference exactly.
- ❑ Zero differing quantized words.
- ❑ Matching quantized-reference and decoded SHA-256.
- ❑ Physical file length equals the reported complete size.
- ❑ Truncation, invalid identifiers, invalid offsets, and checksum errors are rejected.
- ❑ RAW/PACKED_K wins exact ties and protects incompressible tensors.

15. Quality Gates

- ❑ Aggregate held-out quality retention $\geq 95\%$.
- ❑ No critical subgroup falls below its separately defined floor.
- ❑ Calibration and validation data are not reused as the final held-out result.
- ❑ No NaN/Inf activations or unstable generation are introduced.
- ❑ Special/control-token behavior remains correct.
- ❑ Numerical, code, multilingual, and domain-specific subsets are reported separately.
- ❑ The simplest passing policy is preferred over a more complex policy with negligible gain.

16. Go/No-Go Decision Rules

Outcome	Decision	Next action
≤ 8 BPW and $\geq 95\%$ held-out quality	GO	Freeze the policy; complete runtime and subgroup validation.
8.0–8.5 BPW and strong quality margin	PROMISING	Use measured entropy, sparse recovery, and embedding/MLP tiers to close the gap.
Near 9 BPW with $\geq 95\%$ quality	PARTIAL	Quantization works, but post-coding or more K4 allocation is needed.
≤ 8 BPW but $< 95\%$ quality	NO-GO	Restore precision to sensitive families or increase exception coverage.
Post-coding does not beat packed K	REJECT CODEC	Keep the simpler packed representation.
All quantized candidates remain near their fixed K entropy	STOP ENTROPY BRANCH	Do not add further matrix or dictionary sweeps.

17. Deliverables

- Frozen per-tensor or per-region precision map.
- Quantized-reference SafeTensors or equivalent checkpoint.
- Independent encoder and decoder.
- Candidate-by-candidate rate report.
- Exception-map and correction-stream report.
- Exact container round-trip report and hashes.
- Quality/size Pareto curve for 100%, 99%, 98%, 97%, and 95% retention.
- Final complete BPW and physical file size.
- Runtime benchmark and deployment notes.
- Documented negative results for rejected codec families.

18. Recommended First H95Q Run

12. Candidate A: reproduce the existing 7/5 policy and ~9.29 BPW.
13. Candidate B1: move low-sensitivity MLP tensors from K5 to K4.
14. Candidate B2: test embedding K5 while keeping sensitive attention at K6/K7.
15. Candidate C1: use K4 for large robust tensors and K6/K7 sparse recovery for protected channels.
16. Measure retained-symbol entropy and bit-plane entropy before implementing advanced coding.
17. Run rANS only where expected savings exceed table and framing cost.
18. Run quantized matrix/pair coding only where zero residuals or pair coverage are materially elevated.
19. Freeze the simplest policy that approaches 8 BPW with validation margin, then perform held-out quality evaluation.

Conclusion: PBR-H95Q provides a technically defensible path toward approximately 8 total BPW: use mixed K7/K6/K5/K4/K3 precision as the primary reduction, use sparse high-precision recovery to protect quality, and apply matrix, rANS, Huffman, bit-plane, or pair coding only when the quantized data demonstrates measurable structure. The final acceptance condition is both complete serialized rate and held-out model quality—not compression ratio alone.